# Exercise 1.1

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Let $R$ be a commutative ring where $0 \neq 1$. Suppose that every ideal $\left( \neq (1) \right)$ is prime. Show that $R$ is a field.

*Proof.* Clearly $R$ is a domain. Let $a \neq 0 \in R$ be a non-unit. Then $(a^2)$ is a prime ideal and thus $a^2 \in (a^2)$ implies that $a \in (a^2)$. It follows that there exists an element $r \in R$ such that $a = a^2 r$. As $R$ is a domain, we have $1 = ar$ which leads to contradiction.

Let us say that a saturated multiplicatively closed set $S$ is generated by $\{x_i\}$ if $S$ is the smallest such set containing the $x$'s. Prove that if $S$ is finitely generated in this sense, then it can be generated by a single element.

*Proof.* Suppose that $S$ is generated by $x_1, ..., x_n$; we show that $S$ is also generated by $x_1 ... x_n$;denote $S'$ the saturated multiplicatively closed set generated by this single element. $S' \subseteq S$ for $S$ contains $x_1 ... x_n$; $S \subseteq S'$ for $S'$ contains the $x_i$'s. Therefore $S = S'$.